

Abstract: Patient falls during hospital stays are a leading cause of injury, extended recovery, and increased healthcare costs. To address this issue, we have developed SafeStep, a **fall prevention device** using accelerometer and gyroscope data to **detect and respond to real time falls**. Our machine learning model was designed to classify fall events triggering a CO2 based inflation system designed to protect the pelvic region of the patient upon impact.

Introduction

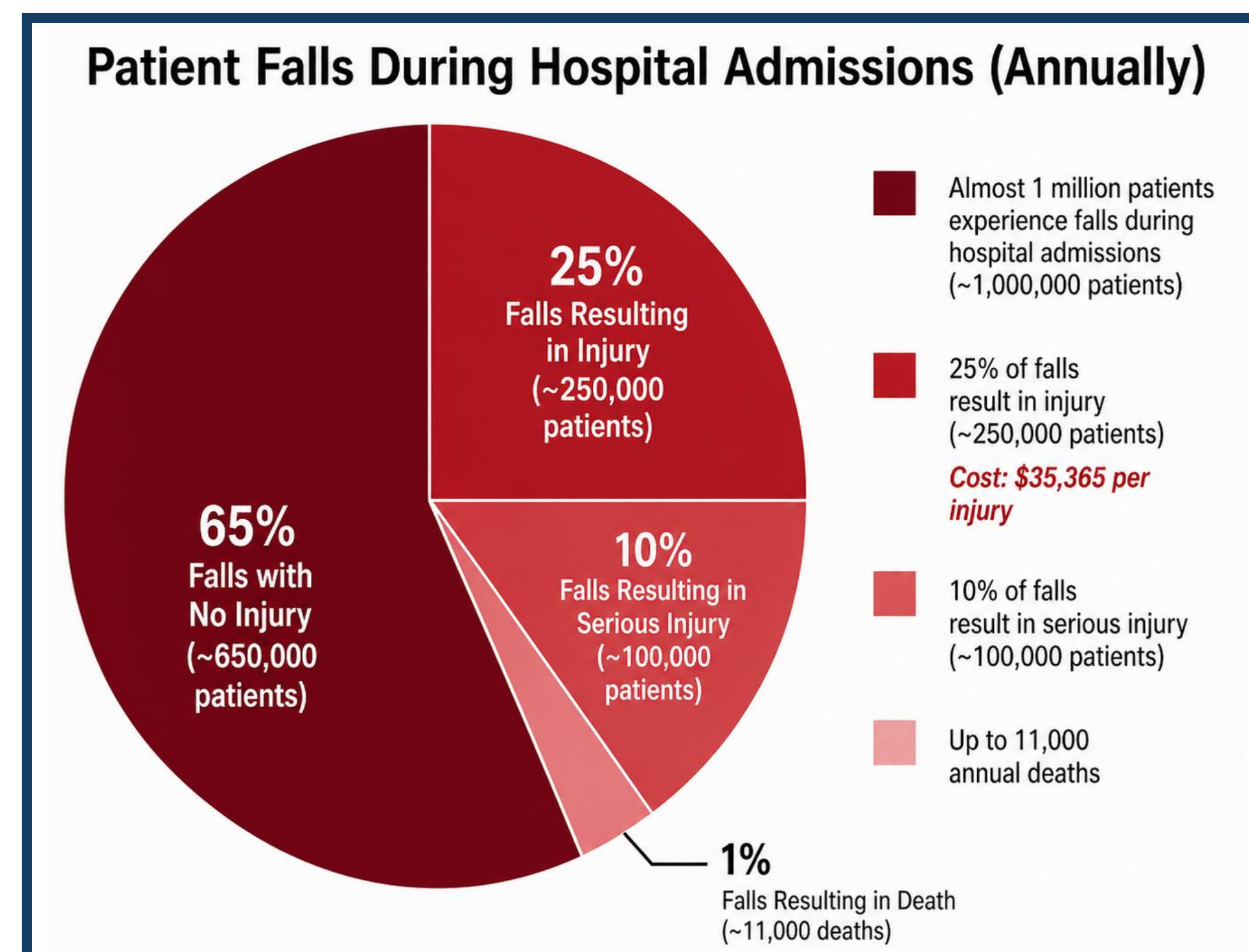


Fig. 1: Statistics of Annual Falls in Hospitals

- Harnesses live sensor data and responds to fall events based on predicted fall windows
- Directly prevents fall in real time
- Mitigates injury to patient by targeting critical area

Proposed System

Utilized an MPU-based sensor only, as ECG leads did not enhance or provide meaningful data to improve fall detection performance.

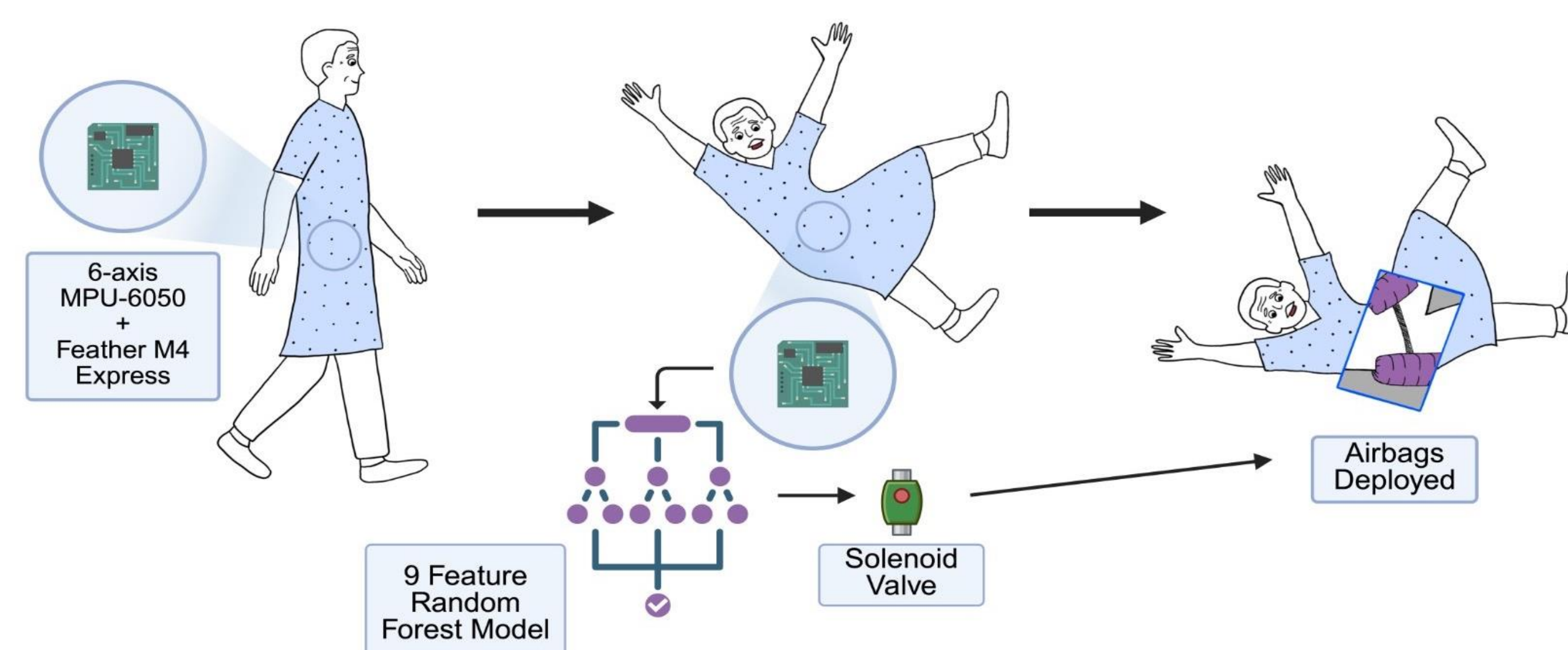


Fig. 2: End-to-End Implementation of a Wearable Fall Prevention System

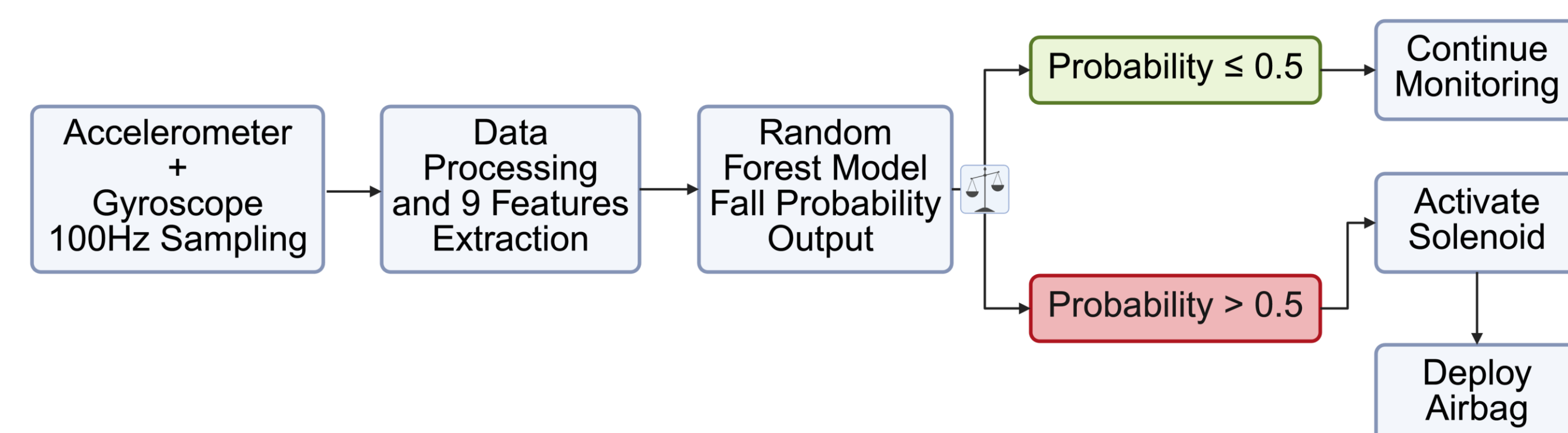


Fig. 3: System Architecture for Real-Time Fall Detection and Airbag Deployment

Prototype Testing

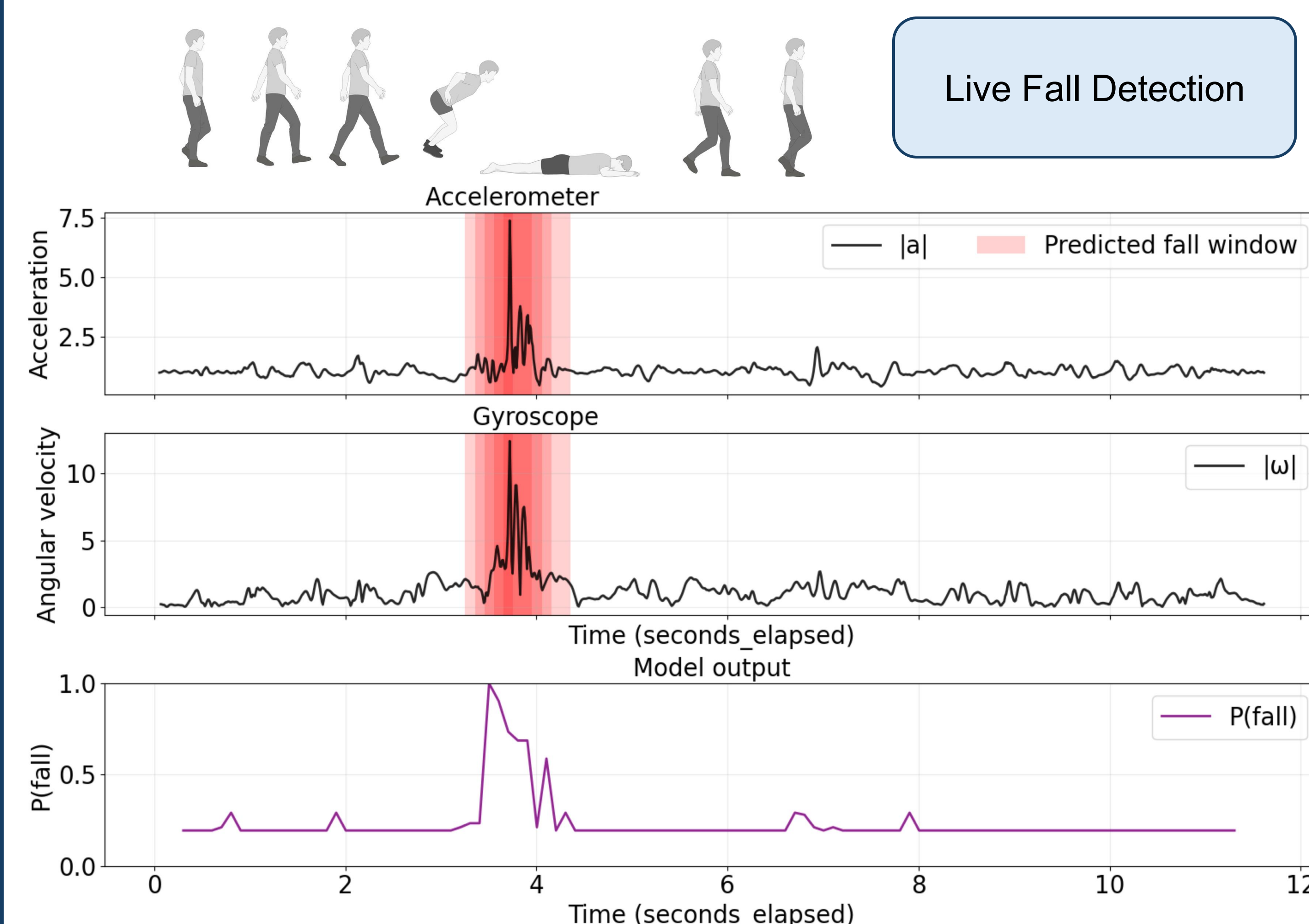


Fig. 4: Live sampling of a fall event. The sensor data depicts real-time readings from the accelerometer and gyroscope over time. Red markers indicate the model's predicted fall window based on live data stream analysis. The airbag system is programmed to deploy when the fall probability exceeds the threshold ($P(\text{fall}) > 0.5$)

Casing Materials:

PLA: Exhibited brittle failure under compression, indicating limited use for impact absorption but adequate rigidity for structural support.
TPU: Demonstrated high elasticity, returning to its original dimensions after compression (28 mm × 29 mm × 29 mm) and withstanding applied forces
Foam: Showed effective force attenuation, reducing transmitted loads during compression testing.

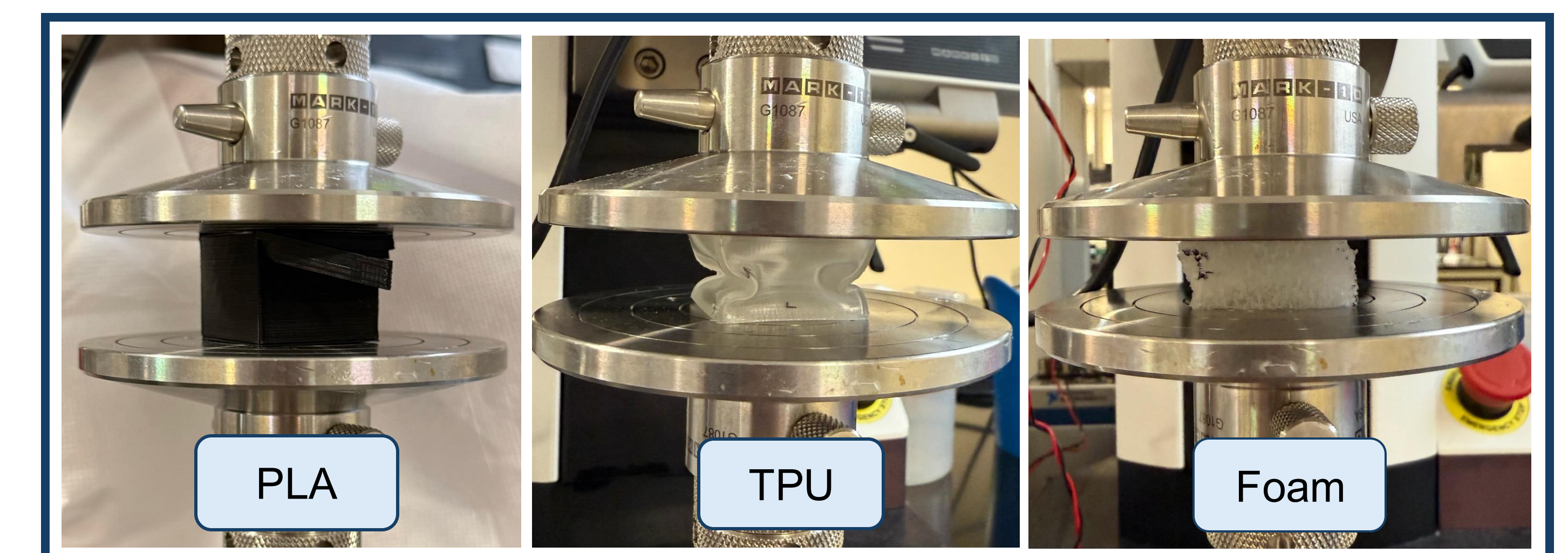


Fig. 5: Compression testing of PLA, TPU, and Foam.

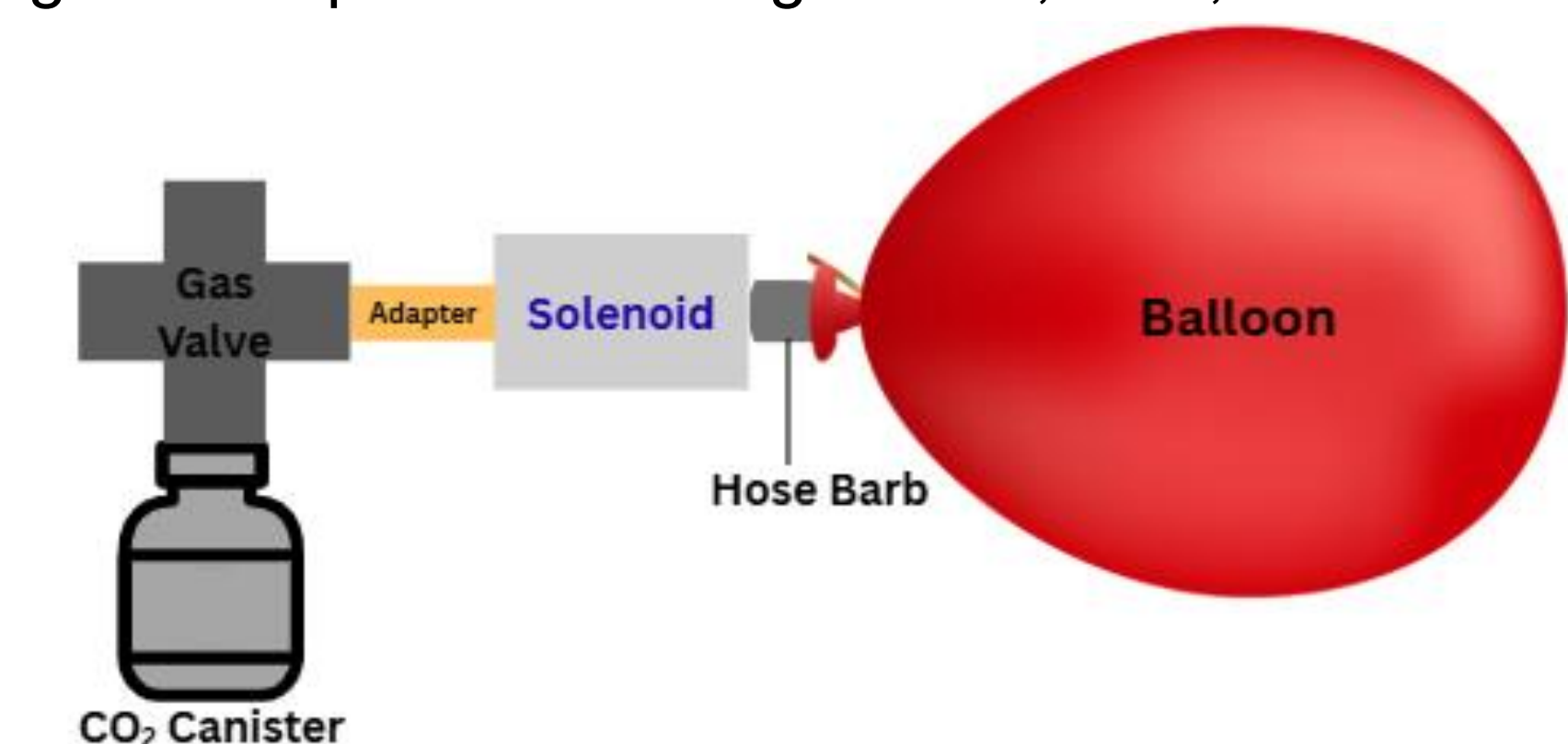


Fig. 6: Inflation System Design. The pad (balloon) is inflated with CO2 gas, supplied by a canister and controlled by a solenoid.

Final Prototype

Due to the absence of in-house test data, model accuracy was evaluated using the Kaggle training dataset, achieving 87%.

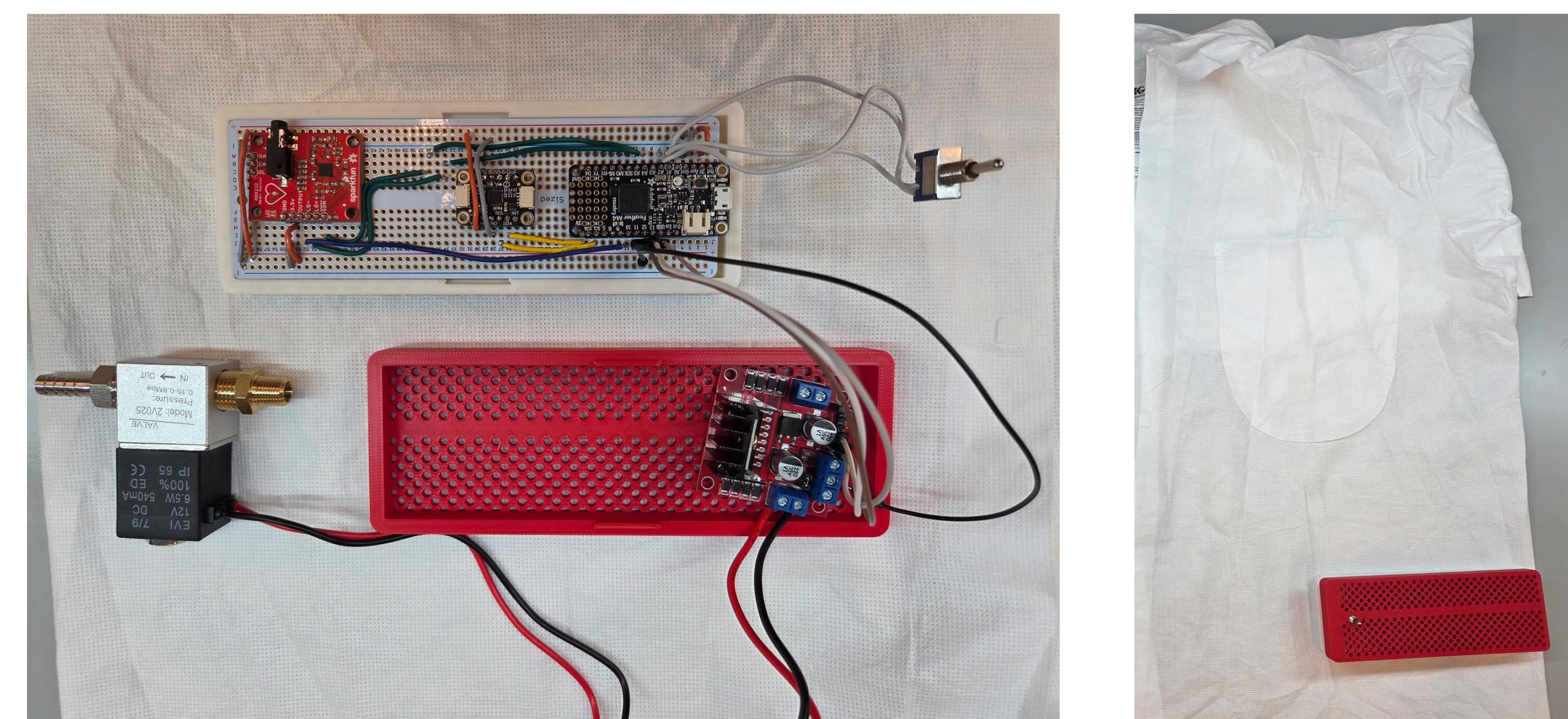


Fig. 7: Circuit system set up with housing compartment; placement of housing compartment on "hospital gown"

Conclusion

Current:

- Fall Detection: gyroscope and accelerometer data collected by the sensor
- Inflation System: Inflates the balloon

Future & Improvements:

- Latency issue with the circuit not classifying a fall and deploying within the timeframe of a fall.
- Classification code to be processed faster,
- Layered casing made up of PLA, foam and TPU
- Experiment with the whole device on a real fall.

References

